Please note that this syllabus is updated on 3/20/2020.

CLASS TIMES AND LOCATIONS

- Math 142-505 : MWF 9:10-10:00am HELD 109

  *As of 3/23/20, the instructor will be available via Zoom during class time.

CATALOG DESCRIPTION AND PREREQUISITES

Business Calculus (Math 1325, Math 1425) Limits and continuity; techniques and applications of derivatives including curve sketching and optimization; techniques and applications of integrals; emphasis on applications in business, economics, and social sciences. Only one of the following will satisfy the requirements for a degree: MATH 131, MATH 142, MATH 147, MATH 151, and MATH 171. Prerequisites: MATH 140 or equivalent or acceptable score on Texas A&M University math placement exam.

LEARNING OUTCOMES

Upon successful completion of this course, students will:

- Logically formulate mathematical variables and equations to quantitatively create mathematical models representing problems in everyday life, as well as business, so that calculus can be applied to achieve an optimal solution.
- Identify patterns in numeric data to calculate limits and derivatives of functions numerically.
- Justify whether a function is continuous or not using the mathematical definition of continuity.
- Understand the derivative as a rate of change in order to quantitatively apply it to everyday life as well as business applications such as marginal analysis.
- Investigate the relationship between a function and its first and second derivatives, and use the information obtained from its derivatives to identify pertinent information about the function.
- Apply the definite integral to quantitatively determine solutions to problems in everyday life and business such as area between curves, average value of a function, and producers' and consumers' surplus.
- Recognize and appreciate the relationship between the derivative (rate of change) and the definite integral (accumulation of change), and utilize the Fundamental Theorem of Calculus as the bridge between the two.

CORE OBJECTIVES

Critical Thinking

- Students will analyze a function and justify whether or not it is continuous using the definition of continuity.
- Students will use inquiry to determine the best method for taking derivatives of complicated functions.
- Students will identify and categorize information about a function in order to construct a graph of its derivative.
- Students will apply calculus to find innovative ways to graph complicated functions without the aid of technology.
- Students will analyze and synthesize data and think creatively to develop mathematical models for optimization purposes.
- Students will examine how the Fundamental Theorem of Calculus connects differential and integral calculus.

Communication Skills

- Students will symbolically relay mathematical information and concepts by creating variables and writing equations.
- Students will recognize, construct, and interpret graphs of basic functions.
- Students will write mathematical information symbolically to describe the behavior of functions.
- Students will justify results that use mathematical definitions such as the definition of continuity by writing proofs.
- Students will explain verbally in class the connection between derivatives, rates of change, and slopes of tangent lines.
- Students will develop sketches of the graphs of complicated functions by analyzing their first and second derivatives.
- Students will explain (both in writing and verbally) mathematical solutions to problems.
- Students will be required to answer questions during lecture concerning topics discussed in class.

Empirical and Quantitative Skills

- Students will evaluate limits numerically and use the information to draw conclusions about the behavior of a function.
- Students will calculate a derivative numerically and explain the result in the context of the problem.
- Students will use marginal analysis to make informed and quantitative business decisions.
- Students will manipulate empirical data to develop a mathematical model to use in an optimization problem, such as maximizing revenue or minimizing cost, and then apply calculus to find and interpret the optimal solution.
- Students will apply the Fundamental Theorem of Calculus to quantitatively compute the accumulated change of a quantity.

INSTRUCTOR INFORMATION

Name	**Arpan Pal**
Email	arpantamu@tamu.edu
Office	Blocker 625X
Office Hours	Mondays and Thursdays 2:30-4:00 in Blocker 625X(my office) *As of 3/23/20, there will be no in person office hours.
Course Page	https://www.math.tamu.edu/~arpantamu/Teaching/Spring2020/142_505_sp2020.html and please login to http://ecampus.tamu.edu
Phone	Math Department: 979-845-3261 (There is no phone in my office, so email is a better way to reach me.)

REQUIRED MATERIALS

TEXTBOOK: *Calculus: Applications and Technology* by Tomastik, 3rd Edition
Note: You will be required to purchase access to the online homework system, WebAssign, but doing so will automatically give you access to the eBook. There are a variety of purchasing options available (course specific access or Cengage Unlimited). This access can be purchased through the local bookstores or on WebAssign. Starting on the first day of classes, you will be granted access for a trial period while you determine the appropriate purchasing option for you.

WEBASSIGN ACCESS: WebAssign will be used for homework in this class. In order to use WebAssign, you must purchase access. For access purchasing information and options, please visit
http://www.math.tamu.edu/courses/eHomework/

CALCULATOR: A TI-83 (any version), TI-84 (any version) or the TI-Nspire (non-CAS version) calculator is **REQUIRED,** and you must bring your calculator to each class. If you need to use a calculator other than those listed, it MUST not perform symbolic mathematics and **you must have my permission to do so**. I will be demonstrating calculator techniques using the TI-84. You must bring your calculator to every class period. You may not share calculators during exams or quizzes.

TEXAS A&M STUDENT ID: Bring your student ID to each class.

ACCESS TO ZOOM: As of 3/23/20, you are required to have access to Zoom. Downloading the client to your computer provides the most seamless experience. To get started, visit https://tamu.zoom.us/download. Click **Download Zoom Client for Meetings**. Once installed, click **Sign In with SSO**. Type **tamu** in the box before .zoom.us. Click **Continue**. Log in with your NetID and password.

TENTATIVE COURSE TOPICS AND CALENDAR OF ACTIVITIES

WEEK	TOPIC	SECTIONS
Week 1: 1/13 – 1/17	Brief Precalculus Review, Limits and Continuity	Review, 3.1
Week 2: 1/20 – 1/24	Limits and Continuity	3.1
Week 3: 1/27 – 1/31	Rates of Change, The Derivative	3.2, 3.3
Week 4: 2/3 – 2/7	Simple Derivative Rules and Marginal Analysis, Product and Quotient Rules	4.1, 4.2
Week 5: 2/10 – 2/14	Review, **EXAM I (3.1-3.3, 4.1, and 4.2)**	
Week 6: 2/17 – 2/21	Chain Rule, Derivatives of Exponential and Logarithmic Functions, Analyzing Graphs with the First Derivative	4.3, 4.4, 5.1
Week 7: 2/24 – 2/28	Analyzing Graphs with the Second Derivative, Limits at Infinity, Curve Sketching Techniques	5.2, 5.3, 5.4
Week 8: 3/2 – 3/6	Absolute Extrema, Optimization	5.5, 5.6
Week 9: 3/9 – 3/13	Spring Break – No Classes	

Week 10: 3/16 – 3/20	No Classes	
Week 11: 3/23 – 3/27	Review, **EXAM II (4.3, 4.4, and 5.1-5.5)**	
Week 12: 3/30 – 4/3	Antiderivatives, Substitution	6.1, 6.2
Week 13: 4/6 – 4/10	Estimating Distance Traveled, The Definite Integral	6.3, 6.4
Week 14: 4/13 – 4/17	Fundamental Theorem of Calculus Part 2 and Average Value of a Function, Area Between Curves, 6.7 topic: Producers' and Consumers' Surplus	6.5,6.6, 6.7topic(optional)
Week 15: 4/20 – 4/24	Review, **EXAM III (5.6 and 6.1-6.6, 6.7 topic(optional))**	
Week 16: 4/27 – 5/1	Review for Final Exam, **Final Exams**	
Week 17: 5/4 – 5/8	**Final Exams**	

Grading Policies

The course grading will be based on the tables below. At the end of the semester, you will receive the grade you *earned* according to the scale given. Due to FERPA privacy issues, I cannot discuss grades over email or phone. If you have a question about your grade, please come see me in person.

As of 3/23/20, the final exam will now be optional. Your grade will be determined based on one of the tables below, depending on whether or not you choose to take the final exam. This decision must be communicated to your instructor by their given deadline which will be a date after you have received your grade for Exam 3.

GRADE BREAKDOWN WITH FINAL

Activity	Date	Percentage
Exam I (3.1-3.3, 4.1-4.2)	2/14/20	20%
Exam II (4.3-4.4, 5.1-5.5)	3/27/20	20%
Exam III (5.6, 6.1-6.6, 6.7 topic (optional))	4/24/20	20%
Computer Homework	Weekly	15%
Quizzes	Weekly	5% *As of 3/23/20, there will be no more quizzes
Final Exam	See below	20%
TOTAL		**100%**

GRADE BREAKDOWN WITHOUT FINAL

Activity	Date	Percentage
Exam I (3.1-3.3, 4.1-4.2)	2/14/20	26.67%
Exam II (4.3-4.4, 5.1-5.5)	3/27/20	26.66%
Exam III (5.6, 6.1-6.6, 6.7 topic (optional))	4/24/20	26.67%
Computer Homework	Weekly	15%
Quizzes	Weekly	5% *As of 3/23/20, there will be no more quizzes
TOTAL		**100%**

Grading Scale

Range	Grade
90 ≤ Average ≤ 100	**A**
80 ≤ Average < 90	**B**
70 ≤ Average < 80	**C**
60 ≤ Average < 70	**D**
Average < 60	**F**

GRADE APPEAL POLICY: Any questions concerning the grading of a Quiz or Exam must be presented to me within one week of the return of the assignment. Otherwise the grade will not be changed. I will be posting grades during the semester on e-campus. Please go to http://ecampus.tamu.edu to login.

HOMEWORK

Graded homework assignments will be completed using your WebAssign account. Additional information regarding online homework:

- Go to http://www.math.tamu.edu/courses/eHomework/ to access your online homework (As well as tutorials for how to use WebAssign).
- You have a "practice" version and a "homework" version for each assignment. There are 20 attempts for each question in the practice version and you have 3 attempts for each question in the homework version (you can submit the answer(s) to each question individually). **The practice versions are NOT counted towards your grade.**
- After submitting an answer in the practice version, you will see the correct answer. **It is very important that you work the practice version at least once, so you will see the format you need to use for your answer in WebAssign.**
- You should use Mozilla Firefox and have the most updated version of Java and Flash on the computer you are using to alleviate technical problems.
- If you have technical issues with WebAssign, please fill out the "Student Help Request Form" found at https://www.math.tamu.edu/courses/eHomework.
- I shall not give an extension due to technical difficulties, so be sure to start your homework well in advance so that you have the time to resolve any technical issue.

CLASS NOTES

On our course page https://www.math.tamu.edu/~arpantamu/Teaching/Spring2020/142_505_sp2020.html you will find a shell and completed notes for each chapter. *As of 3/23/20, the plan is to post videos that you can watch on your own time to complete the class notes.

HIGHLY SUGGESTED HOMEWORK

Your daily highly suggested homework will prepare you for your computer homework, quizzes, and exams, but will not be turned in for a grade. It is crucial that you work these problems. A list of highly suggested homework problems can be found at the end of this syllabus and on the course webpage and in eCampus.

QUIZZES *AS OF 3/23/20, THERE WILL BE NO MORE QUIZZES. YOUR CURRENT QUIZ AVERAGE WILL REPRESENT 5% OF YOUR FINAL GRADE.

There will be weekly quizzes throughout the semester.

EXAMS

There will be **three in class exams** during the semester. You must bring your student id, approved calculator, and grey TAMU Scantron to each exam. Calculators will be checked before or during each exam. If there are any programs, notes, or formulas on your calculator which I did not give you, the occurrence will be considered scholastic dishonesty. The tentative exam schedule is as follows:

Exam I: Friday, February 14, 2020 **Exam II**: Friday, March 27, 2020 **Exam III**: Friday, April 24, 2020

*As of 3/23/20, Exam II and Exam III and the Final Exam will be given online. Further details will be disseminated as they are determined.

FINAL EXAM

The final exam will be **comprehensive** and is NOT **required** for all students. If your final exam grade is higher than your lowest test grade, the grade on your final will replace that test grade in the final grade calculation. The final exam schedule is as follows:

Section	Class Time	Tentative Final Exam Date, Time, ~~and Location~~
142-505	MWF 9:10-10:00am	Friday, May 1st, 2020, 8:00-10:00am in ~~HELD 109~~

(You can refer to http://registrar.tamu.edu/Courses,-Registration,-Scheduling/Final-Examination-Schedules for the University final exam schedule.)

ATTENDANCE AND MAKE-UP POLICIES

Attendance is essential to complete this course successfully.

- **Excused Absences**: University student rules concerning excused and unexcused absences, as well as makeups, can be found at http://student-rules.tamu.edu/rule07. In particular, make-up exams and quizzes or late homework will NOT be allowed unless a **University approved reason is given to me in writing**. Notification *before* the absence is **required** when possible. Otherwise (e.g. accident, or emergency), you must notify me **within 2 working days** of the missed exam, quiz, or assignment to arrange a makeup. In all cases where an exam/quiz/assignment is missed due to an injury or illness, whether it be more or less than 3 days, **I require a doctor's note**. I will not accept the "University Explanatory Statement for Absence from Class" form. Further, an absence due to a non-acute medical service or appointment (such as a regular checkup) is *not* an excused absence.
- If you have a University approved absence for missing an exam, you will be expected to make up your exam according to the Department Make-up Schedule that can be found at http://www.math.tamu.edu/courses/makeupexams.html, starting with the first option for each exam. Only if you have a University approved absence for the day of the exam and the previous makeup day will you be allowed to use the later options or have other arrangements made. You must discuss (email is fine) the need for a make-up exam with me before going to a scheduled time. *As of 3/23/20, it is unknown how make-ups will be handled. If you are excused from an exam, please notify the instructor as soon as possible so that arrangements for a make-up exam can be made.

ACADEMIC INTEGRITY

"An Aggie does not lie, cheat, or steal, or tolerate those who do."

Upon accepting admission to Texas A&M University, a student immediately assumes a commitment to uphold the Honor Code, to accept responsibility for learning, and to follow the philosophy and rules of the Honor System. Students will be required to state their commitment on examinations, research papers, and other academic work. Ignorance of the rules does not exclude any member of the TAMU community from the requirements or the processes of the Honor System. For additional information please visit http://aggiehonor.tamu.edu.

AMERICANS WITH DISABILITIES ACT (ADA)

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources in the Student Services Building or at (979) 845-1637 or visit http://disability.tamu.edu. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

ADDITIONAL CLASS POLICIES

- **Email Policy:** Check your official TAMU email regularly. Please include your full name, course number (142), and section number (505) in your emails. If any of this information is missing, it may delay my response.
- **Electronic Device Policy:** Calculators are the only electronic device permitted during class. If I hear or see a student using other devices, they may be asked to leave.
- **Positive Environment:** It is my responsibility to create a learning environment in which everyone respects one another and their fellow Aggie's opportunity to learn. You can help me create this learning environment by being on time to the class, staying focused on the math concepts we are talking about (no electronic device other than your calculator), and not leaving early. I hope that this idea of respectfulness comes naturally to all Aggies but if it ever becomes a problem you may be asked to leave the classroom.

AVAILABLE HELP

If at any point in the semester you are not fully understanding the concepts we are covering, please seek help immediately (not 2 months later). Please keep in mind all of the resources available to you:

- *Me (your instructor)* – I want you to succeed in this class! Feel free to ask questions before, during, or after class. If you still have questions feel free to stop by anytime during my office hours. *As of 3/23/20, I will be available to answer questions via Zoom during our regular class time.

- *Help Sessions* - Help sessions are an opportunity for you to ask questions and get help with your homework. These sessions are led by students, where you may come and go, as your schedule allows. Once determined, the schedule will be announced in class, posted on our course webpage, and additionally posted at http://www.math.tamu.edu/courses/helpsessions.html. *As of 3/23/20, help sessions will be moved to Zoom. More details will be disseminated as they are known.

- *Week-in-Review* – There will be Week-in-Review sessions conducted by Mrs. Angie Allen each week, starting the second week of classes. Each review is open to all Math 142 students to review the topics of the previous week and to provide additional examples. The schedule and problem sets that will be worked during these sessions can be found at https://www.math.tamu.edu/~aallen/142spring2020wir.html*As of 3/23/20, there will be no in person week-in-reviews. A live week-in-review via Zoom will be held during the originally scheduled times.

TITLE IX AND STATEMENT ON LIMITS TO CONFIDENTIALITY

Texas A&M University and the College of Science are committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws provide guidance for achieving such an environment. Although class materials are generally considered confidential pursuant to student record policies and laws, University employees — including instructors — cannot maintain confidentiality when it conflicts with their responsibility to report certain issues that jeopardize the health and safety of our community. As the instructor, I must report (per Texas A&M System Regulation 08.01.01) the following information to other University offices if you share it with me, even if you do not want the disclosed information to be shared:

- Allegations of sexual assault, sexual discrimination, or sexual harassment when they involve TAMU students, faculty, or staff, or third parties visiting campus.

These reports may trigger contact from a campus official who will want to talk with you about the incident that you have shared. In many cases, it will be your decision whether or not you wish to speak with that individual. If you would like to talk about these

events in a more confidential setting, you are encouraged to make an appointment with the Student Counseling Service (https://scs.tamu.edu/).

Students and faculty can report non-emergency behavior that causes them to be concerned at http://tellsomebody.tamu.edu

HIGHLY SUGGESTED HOMEWORK PROBLEMS

Section	Problems
EXAM 1	
Review	See my webpage
3.1	1-45(odd), 49, 51, 61-81(odd)
3.2	1-43(odd), 51, 57, 59, 65, 67, 69
3.3	1-13(odd), 23-33(odd), 37-43(odd), 47, 53, 55
4.1	1-75(odd), 79, 97
4.2	1-35(odd), 41, 43
EXAM 2	
4.3	1-37(odd), 41, 45
4.4	1-73(odd)
5.1	1-41(odd), 45-53(odd), 61, 63, 73, 75, 87, 89
5.2	1-35(odd), 41-45(odd), 57, 63, 71, 73
5.3	1-17(odd), 21, 27, 31
5.4	1-35(odd), 41, 47, 49
5.5	1-29(odd), 43
EXAM 3	
5.6	1-9(odd), 13, 15
6.1	1-47(odd)
6.2	1-45(odd)
6.3	1-9(odd), 17-23(odd)
6.4	1-11(odd, n=10 only), 23-29(odd), 33-41(odd), 67
6.5	1-51(odd), 55-63(odd)
6.6	1-35(odd), 41, 51, 59, 61
6.7 (OPTIONAL)	15-25(odd)